Pixels, Frames & Transformers

Bias, Measurement and Faces · Module 07

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Errors That Aren't Random

Gender Shades (Buolamwini & Gebru, 2018) audited commercial gender classifiers on a set of faces balanced by skin type and gender.

- Two benchmarks widely used for face analysis were about 80% and 86% lighter-skinned
- The classifiers' error rates ranged from under 1% for lighter-skinned men to as much as 35% for darker-skinned women

The lesson for any vision model: a classifier that is accurate on average can be much worse for some groups, usually the ones that were rare in its training data.

Why social scientists should care: if errors differ between the groups you compare, your comparison is biased (slide 8).

Who Is in the Data, and Who Labeled It?

FairFace (Kärkkäinen & Joo, 2021)

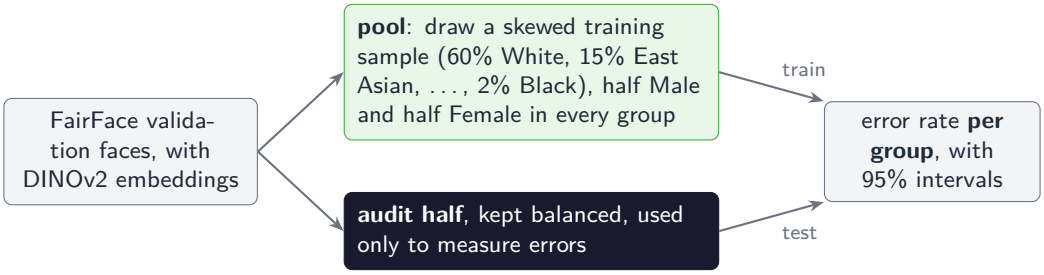
- About 108,000 faces, mostly from public Flickr photos
- Collected until seven race groups were roughly balanced, also by gender and age
- Labels from **crowd workers** who looked at each face

CC BY 4.0. Built for exactly what we need: checking performance *per group*.

What the labels are, and are not
Gender labels are binary and record what annotators *perceived*. Race labels are annotators' guesses at one particular scheme of social categories. Neither says how a person identifies.

In this module we train a gender classifier only to audit it. Don't use such a model to label real people.

Audit Design: Train Skewed, Test Balanced



The skew is in *who appears*, not in the outcome. Then repeat with a balanced training sample of the same size: does the gap close? Often it narrows; it may not vanish, because the pre-trained embeddings carry their own training data's skew.

Report Error Rates per Group, with Intervals

- One overall accuracy hides how errors are distributed
- Report the error rate **per group**, and per **intersection** (race × gender): Gender Shades found the largest errors there
- Use the **bootstrap**: resample each group's faces many times and see how much its error rate moves

How precise is an error rate of 10%?

faces in the group	95% interval
50	± 8.3 points
200	± 4.2 points
800	± 2.1 points

To tell a gap of a few points from noise, you need hundreds of faces *per subgroup*.

What Do Embeddings Encode About People?

A **probe**: try to predict an attribute from the embeddings with cross-validation. If it works, the information is there, and **any** model trained on the embeddings can use it.

On the Congress portraits, we probe for attributes from the public metadata (never inferred from faces): party, gender, chamber, and whether a member entered Congress in 2017 or later.

Two kinds of information
about the *people* (appearance correlated with gender or age), and about the *photos* (backdrop, lighting, the year it was taken, the chamber's studio).

Shortcut learning: a model trained to code "formal versus casual" photos may quietly learn gender or photo era instead. Audit it across those attributes.

Categories, Skin Tone and Identity

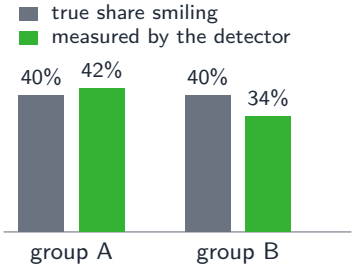
Annotated categories
someone's perception of a face, in someone's scheme of categories (FairFace's seven groups)

Skin tone scales
appearance, not identity: Fitzpatrick (6 types) or the 10-point **Monk Skin Tone** scale (Google, 2022), with more detail for darker tones

Self-report
measures identity directly, with consent, but rarely exists for images from the web

Inferring gender, race, age or emotion from a face is a measurement problem as much as an ethical one. Alternatives: use metadata or self-reports; code the *context* (role, framing) instead of identity; or have trained coders code perceived attributes and report agreement.

When Errors Differ by Group, Conclusions Shift



expected values in the simulation

A smile detector finds 90% of smiles in group A but 70% in group B, with 10% false alarms in both. It is right 86% of the time, and it **invents** an 8-point gap.

Fix: a validation sample per group. Hand-code a random sample, estimate each group's sensitivity and false positive rate, and correct (Rogan & Gladen, 1978):

$$\text{corrected} = \frac{\text{measured} - \text{false positive rate}}{\text{sensitivity} - \text{false positive rate}}$$

Regression methods that carry this uncertainty through: TeBlunthuis et al. (2024).

What to Report

When a measurement in your study involves people in images, readers need to know:

1. **Who is in your data**, and how that compares with the population you make claims about
2. **How labels about people were obtained**: metadata, self-report, or someone's perception, and in which categories
3. **Error rates per relevant group**, with intervals and the size of each group in your validation sample
4. **What you did about differences**: rebalancing, correction, or a stated limitation
5. **What you chose not to infer**, and why

In the Notebook

- FairFace: who is in it, and how it was labeled
- A Gender Shades-style audit: skewed versus balanced training data
- Error rates per group and per intersection, with bootstrap intervals
- Probing portrait embeddings for people and for photo production
- Skin tone, categories and identity
- A simulation of differential error, and a correction with a validation sample

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07_bias_measurement_and_faces_exercises.ipynb · 5 exercises · GPU recommended